PHOTO FRAME GS 02

E-Ink Display 960x540px [LilyGo T5 E-paper Plus / ESP32 S3]

E-ink digital picture frame with SD + LittleFS image storage and remote management via FTP/Telnet (including HTTP(S) fetch), plus built-in file operations (list/copy/delete). Features dual-slot OTA firmware updates, protected Telnet authentication (session timeout + progressive lockout), and ultra-low-power deep sleep with RTC backup and scheduled wake-up hour for long battery life.

CONNECTION & ACCESS

WIFI AP MODE *(Default)*

SSID: PhotoFrameGS02
Password: 123456789
IP: 192.168.4.1
Gateway: 192.168.4.1
Subnet: 255.255.255.0

WIFI STA MODE

sta_ssid – WiFi name
sta_password – WiFi password
sta_enable – static IP
sta_ip/gateway/subnet/dns
Set: ap_enable = false

mDNS (optional)

mdns_enable = true
mdns_hostname = photoframegs02
Access: photoframegs02.local

TELNET ACCESS

Port: 23
User: admin
Pass: 123456789
Session: 1 hour

FTP ACCESS

Port: 21
User: admin
Pass: 123456789

TELNET COMMANDS

SYSTEM

help – command list
clear – clear screen
reboot – restart device (confirm)
reset config – factory reset (confirm)
exit – disconnect
logout – re-auth (confirm)

INFO

imginfo – current image
meminfo – RAM/PSRAM
netinfo – IP/MAC/DNS
fsinfo – SD/LFS usage
sketchinfo – FW/sketch info
nvsinfo – NVS usage
batinfo – battery status
date – system time
date rtc – RTC time
date rtc set YYYY.MM.DD HH:MM:SS – set RTC
date rtc sync-from-ntp – RTC <– NTP
date rtc sync-to-system – SYS <– RTC
timestamp – Unix epoch

FILES

list – list active storage
list sd|sdcard – list SD
list lfs|littlefs|fallback – list LittleFS
list logs – list logs tree
cat <filename> – show file
copy sd lfs <file> – SD -> LFS
copy lfs sd <file> – LFS -> SD
rename sd <src> <dst> – rename on SD
rename lfs <src> <dst> – rename on LFS
delete sd <file> – delete from SD *(confirm)*
delete lfs <file> – delete from LFS *(confirm)*
format sd|lfs – format storage *(confirm)*

CONFIG

config <key> – get value
config <key> <value> – set value

DOWNLOAD / UPDATE

fetch <url> [name] – HTTP/HTTPS, max 400KB, .jpg/.jpeg
fwupdate – update status
fwupdate verify – verify firmware
fwupdate run – apply update *(confirm)*
bootpart status|ota0|ota1 – boot slot info/set

FTP & CONFIGURATION

FILE STRUCTURE

/config.ini – settings
/images/ – photos *(active storage: SD or LittleFS)*
/firmware/ – OTA bundles *(firmware.bin + firmware.sha256)*
/logs/ – boot/status events *(ex. 2026/01/01/20260101.log)*

IMAGE REQUIREMENTS

Format: JPG/JPEG
Size: 960x540px
Max.: 400 KB per file
Tip: grayscale + contrast optimization

```
magick pic01.jpg -colorspace Gray -contrast-stretch 1%x1% -ordered-dither o8x8,16 pic01.jpg
```

CONFIG.INI SECTIONS

[device] – name, version
[display] – jpg settings
[ap mode] – AP settings
[sta mode] – WiFi creds
[static ip] – IP settings
[ntp] – time server
[telnet] – telnet config
[ftp] – ftp config
[timer] – wake interval
[tone] – tone enable *(true/false)*
[storage] – fallback *(enable/disable)*

WAKE UP VALUES

1=Min, **2**=Hour, **3**=12h, **4**=Day, **5**=Week, **6**=Month, **wake_up_hour**=0..23 *(used in Day/Week/Month modes)*

BUTTONS

Maintenance/Wake *(GPIO21)*: hold 3s during boot –> Maintenance mode
Maintenance/Wake *(GPIO21)*: hold 3s in Maintenance –> Exit & reboot
Factory Reset *(GPIO48)*: hold 30s –> Full reset

OPERATING MODES

1 PHOTO FRAME MODE *(Normal Operation)*

- Automatically displays JPG images from the active **/images/** folder *(SD or LittleFS)*
- Uses storage fallback if the primary storage has no images
- Cycles images by configured wake mode: Seconds, Minutes, Hourly, Half-Day, Daily, Weekly, Monthly
- Supports wake_up_hour scheduling (0–23) for Daily/Weekly/Monthly wake modes
- Uses RTC-backed timing for accurate scheduled wake-ups
- Deep sleep between refresh cycles for battery conservation
- Low-battery detection with visual warning and power-safe behavior
- Optional tone alerts *(if enabled)*

2 MAINTENANCE MODE *(Configuration & File Management)*

Activation: Hold Maintenance Button *(GPIO 21)* for 3 seconds during boot
Features available:
- WiFi connectivity *(AP or STA mode)*
- Telnet server for remote configuration and file operations *(list/copy/delete/fetch)*
- Telnet auth protection *(session timeout + lockout)*
- FTP server for file upload/download
- mDNS support for easy hostname access
- OTA maintenance commands *(firmware verify/run, boot slot control)*
- *Boot/status events can be logged*

Exit: Hold Maintenance Button for 3 seconds to reboot
Factory Reset: Hold Reset Button *(GPIO 48)* for 30 seconds

3 LOW BATTERY MODE *(Empty battery)*

- Triggered when battery voltage drops below the low-battery threshold
- Shows battery status on display *(voltage + percentage)*
- Enters extended deep sleep to preserve remaining energy
- Wakes periodically for battery re-check and safe recovery handling
- Low-battery warning tone *(if enabled)*

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